

Is It Possible to Preserve Vision without Compromising Metastases-Free Survival by Use of Fully Fractionated Stereotactic Radiotherapy for Posterior Choroidal Melanoma?

Claire E. Phillips^{a,b} Arkan Youssef^a Mathias Bressel^{a,b} Rod O'Day^c
Joseph Sia^{a,b} John D. McKenzie^c Daniel McKay^c William Campbell^{b,c}
Fred Kuanfu Chen^{c,d}

^aDepartment of Radiation Oncology, Peter MacCallum Cancer Centre, Melbourne, VIC, Australia; ^bThe Sir Peter MacCallum Department of Oncology, University of Melbourne, Parkville, VIC, Australia; ^cDepartment of Ocular Oncology, Royal Victorian Eye and Ear Hospital, East Melbourne, VIC, Australia; ^dLions Eye Institute, The University of Western Australia Nedlands, Nedlands, WA, Australia

Keywords

Posterior choroidal melanoma · Radiotherapy fractionation · Visual acuity

Abstract

Introduction: Stereotactic radiotherapy (SRT) is used for choroidal melanoma (CM) abutting the optic nerve. Visual acuity (VA) deterioration to $\leq 6/60$ is common. We report a pilot study of reduced-dose SRT using 2 Gy/day, aiming to preserve vision without compromising survival. **Method:** 60 Gy SRT was delivered in 30 fractions over 6 weeks. Liver metastasis surveillance was annual ultrasound. The primary endpoint was 5-year metastasis-free survival (5yMFS). Secondary endpoints were 2-year freedom from local progression (2yFFLP), VA, enucleation rate, and radiation toxicity. **Results:** Twenty adults aged ≤ 70 years with T1-T2M0 CM without diabetes mellitus were enrolled. Median follow-up was 5.1 years. About 85% and 90% of tumours were ≤ 3 mm of the macula and optic disc, respectively. Median tumour height was 2.2 mm (range 1.0–4.4 mm), and median basal diameter was 8.2 mm (range: 4.3–15.0 mm).

5yMFS was 88% (95% CI: 61–97), and the 2yFFLP rate was 90% (95% CI: 66–97). There were three enucleations for disease progression. Final VA in retained eyes was $\geq 6/7.5$ in 6 (30%), 6/9 to 6/12 in 5 (25%), 6/15 to 6/48 in 2 (10%), and $\leq 6/60$ in 4 (20%) eyes. Retinopathy was the main cause of vision loss besides tumour progression. **Conclusion:** Meaningful vision was preserved 5 years after SRT, despite high-risk tumour locations for vision loss. 2yFFLP and 5yMFS were acceptable. This dose fractionation warrants further investigation.

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Introduction

Stereotactic radiotherapy (SRT) is an alternative to proton beam therapy (PBT) for choroidal melanoma (CM) in close proximity to the optic nerve (ON) [1]. SRT dose fractionation regimens such as 50–70 Gy in 4 or 5 treatments (fractions) over 1 week were adopted from PBT. These regimens use large fractions which are ablative for tumour and for normal tissues. Local tumour control in the eye is $>90\%$, but deterioration of visual

acuity (VA) to $<6/60$ is expected when the treatment volume includes the ON [2].

One mechanism for reducing late radiation toxicity is to use standard fractionation (1.8–2 Gy/day). This reduces late radiation injury by allowing DNA repair between fractions [3]. There is however a dose threshold above which repair cannot overcome injury, even with standard fractionation [4]. A steep increase in the probability of permanent vision loss from 0% to over 15% is observed after ON doses from 50 to 70 Gy (2 Gy/day). For a single fraction, this steep dose response occurs at only 8–10 Gy. Similarly, the risk of retinopathy steadily increases from a threshold of approximately 50 Gy (2 Gy/fraction) [4].

A second mechanism to reduce toxicity is to reduce the prescribed dose. A wide range of doses are currently used across PBT, brachytherapy, SRT, and stereotactic radiosurgery (single fraction, SRS) with similar tumour control [1]. Given the high rates of local control after ablative doses, several studies have explored dose reduction to reduce late radiation toxicity. Large daily fractions were used in these studies. While disease control was maintained and mild toxicity reduction was observed, vision outcomes were not improved [5, 6].

This pilot study combined a reduced equieffective dose (EQD) [7] with standard fractionation SRT to explore if vision could be preserved without compromising survival. The protocol dose of 60 Gy in 30 fractions is a common treatment for many cancers including lung cancer and brain tumours. Existing data could predict a reduction in rates of ocular toxicities with this dose fractionation, but the tumour and vision outcomes were unknown.

Method

Patients

Inclusion criteria were previously untreated posterior CM, age ≤ 70 years, affected eye VA $>6/60$, tumour height of 2.5 mm–8 mm or observed growth if <2.5 mm, maximum basal diameter of 16 mm, no liver metastases on liver ultrasound (US). Exclusion criteria were involvement of iris or ciliary body, diabetes mellitus, non-malignant eye disease with potential to affect vision, prior cancer with evidence of/or risk of recurrence $>10\%$.

Design

This was a prospective phase II study. The study was conducted according to the Good Clinical Practice Guidelines. Institutional Ethics Committee approval was obtained, and all participants gave written informed consent before any study procedures. All patients received SRT to 60 Gy in 30 fractions over 6 weeks. A thermoplastic mask and light/camera for gaze fixation and tracking were

utilised for treatment simulation and delivery [8]. A high-resolution T2 sequence planning MR orbit was obtained in a thermoplastic mask and the eye tracker-defined gaze position. The MRI defined the tumour volume supplemented by fundus photography and beta US. A 2- to 3-mm expansion margin of the tumour volume gave the treatment volume. Relevant normal structures were contoured for dose-volume analysis (Fig. 1) The retina was artificially defined as a 1-mm thick volume to facilitate more robust dose-volume calculations and comparison between cases. Dose-to-volume limits were placed on the ipsilateral lens and lacrimal gland and the contralateral eye and ON. The maximum allowable dose to the ipsilateral lens was 7 Gy over 30 fractions, which is well below 12 Gy in 6–8 fractions, the fractionated dose that causes certain cataract [4].

Study Assessments

Endpoints

The primary endpoint was 5-year metastasis-free survival (5yMFS). Secondary endpoints were 2-year freedom from local progression (2yFFLP), enucleation rate, best-corrected VA (BCVA), and late radiation toxicity as defined by the Common Toxicity Criteria v4.0 (CTCv4) [9]. Assessments were at baseline and 3, 6, 9, 12 months and then annually. The primary endpoint was assessed with annual liver US and other imaging when clinically indicated.

When this pilot study was designed, it was not known if 60 Gy in 30 fractions would be a sufficient biologic dose for local control of any tumour. An interim analysis was not possible in the statistical design of this small study, but at a pragmatic level, 2-year FFLP was chosen to allow early detection of a clinically unacceptable local control rate (leading to consideration of early closure of the study if necessary).

The Common Toxicity Criteria score treatment toxicities according to both the functional impact of the toxicity and the need for therapeutic intervention, for example, the need for anti-VEGF therapy. The presence or absence of toxicity was scored by the ophthalmologist at clinical examination. The CTC v4.0 grade was then determined at time of analysis by considering the recorded VA at that visit and any need for treatment. No attempt was made to determine how much a particular toxicity contributed to the VA, which may have artificially increased the CTC grade recorded for that toxicity as more than one factor can affect vision.

Statistical Method

Baseline and treatment characteristics were summarised using descriptive statistics. Categorical data were summarised by counts and percentages, and continuous

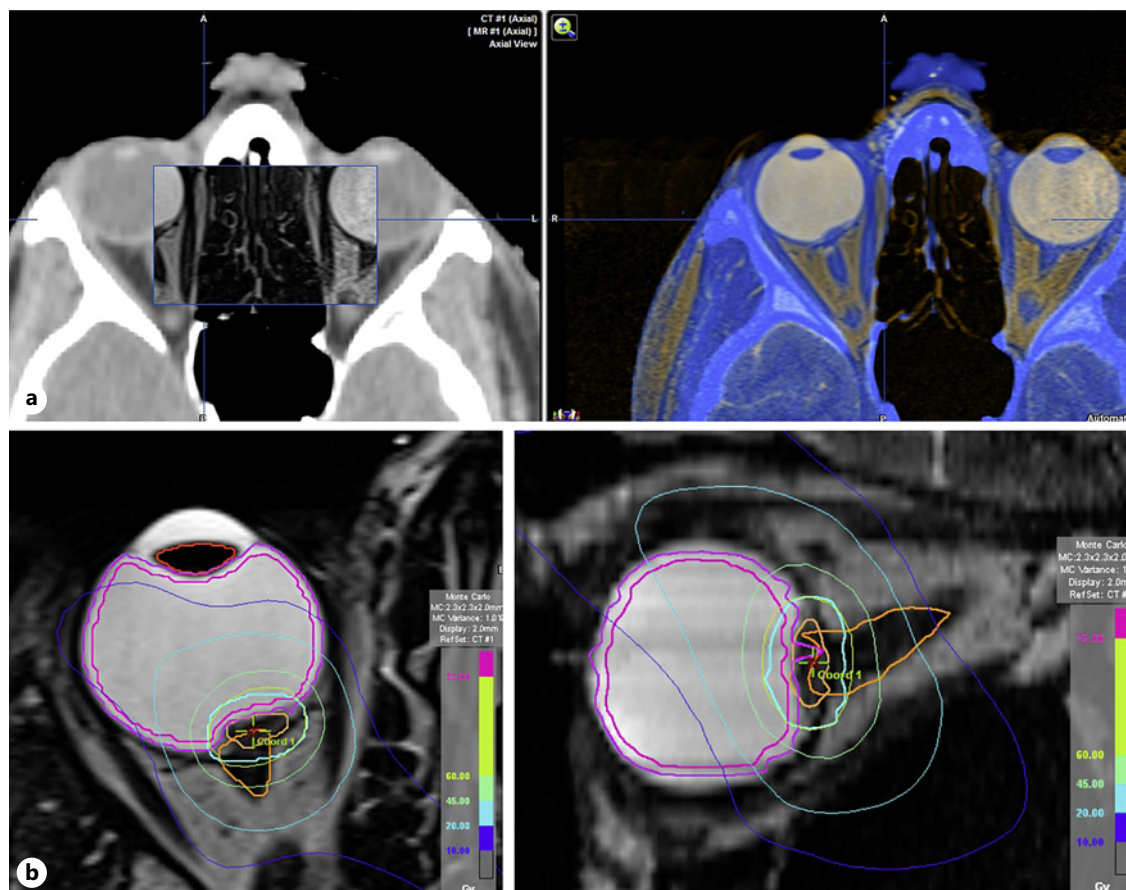


Fig. 1. SRT treatment plan. **a** CT to MRI fusion. **b** Sample treatment plan.

data were summarised by mean, standard deviation, range, median, and interquartile range.

Late radiotherapy toxicity was recorded and described. VA was described over time using a swimmer plot. A swimmer plot was provided to describe the time until local progression, distant progression, enucleation, and death [10]. All statistical analyses were performed in R version 4.2.1 (2022-06-23 ucrt) [11].

Results

Twenty patients were recruited from December 2014 to May 2020. Median age was 47.5 years (range 19–65). Mean height was 2.4 mm, and mean basal diameter 8.3 mm. Ninety percent of tumours were within 3 mm of the ON, and 85% were within 3 mm of the macula. Tumour biopsies were not able to be obtained. Individual patient data are presented in Table 1. Median follow-up was 5.1 years. One patient moved internationally and underwent follow-up at a

UK specialist eye centre. Attendance was not always per protocol from the onset of the COVID-19 pandemic to the end of 2022. All disease and VA data generated or analysed during this study are included in this article. A later manuscript will outline details of the radiotherapy dose-volume analysis (for a radiation oncology audience). Further enquiries can be directed to the corresponding author.

Disease and enucleation events are demonstrated in Figure 2. 5yMFS was 88% (95% CI: 61–97), and the 2yFFLP rate was 90% (95% CI: 68–99). After a median follow-up of 5.1 years, tumour has progressed in three eyes (5yFFLP: 85%, 95% CI: 60–96). Distant metastasis (DM) only occurred with or after disease recurrence in the eye. Three eyes were enucleated for progression. Two of the resected tumours demonstrated loss of chromosome 3 and/or BAP1 mutation. The patient with the third locally progressive tumour refused any therapy until 18 months after initial progression at which time the tumour was re-irradiated with SRT 50 Gy in 5 fractions. At 55 months, the eye was then enucleated after an

Table 1. Individual patient data: baseline characteristics, VA, and radiation toxicity outcomes

Case	Follow-up, years	Tumour size, mm		Distance, mm		VA		CTCv4.0 worst grade		Other events affecting vision
		height	largest basal	ON	macula	baseline	last	retinopathy	ONp	
1	5	1.42	8.33	3	0	N	N	1 resolved	0	
2	5	2.5	10.00	1.5	9	6/12	6/24	3	0	
3	PD	1.00	15.00	0	0	N	–	–	–	Enucleation
4	5	1.30	6.00	0	2	N	N	1 resolved	0	
5	5	2.18	4.43	0	0	N	6/12	0	0	
6	5	2.15	5.78	0.5	1	N	N	1 resolved	0	
7	5	1.68	8.26	3	1	N	N	2	1 resolved	
8	PD	3.93	10.07	1	0.5	6/12	–	–	–	Enucleation
9	5	1.92	7.59	0	1	N	N	0	0	
10	5	2.17	5.84	0	>3	N	6/9	1	1 resolved	
11	5	2.72	7.65	0	1	N	N	2	1	Early cataract
12	PD	4.38	8.00	0	>3	N	–	–	–	Enucleation
13	5	3.50	8.50	0	0	6/15	<6/60	0	–	Outer retinal atrophy
14	5	2.57	10.16	0	0	6/9	<6/60	3	0	
15	4	2.44	8.70	0	0	N	6/12	1	0	
16	4	2.32	8.22	0	0	6/18	<6/60	4	Not available	
17	3	3.26	9.18	1	1	N	6/12	0	0	
18	3	1.77	6.14	1	1	N	6/24	1	0	
19	3	2.32	12	0	0	6/9	N	0	0	
20	3	2.15	6.32	1	0	6/12	<0.10	4	4	Central retinal vein occlusion

ON, optic nerve; N, normal (6/7.5 or better); PD, progressive disease in the eye.

observed increase in tumour height; however, histopathology demonstrated a scar with scant tumour cells, insufficient for chromosome or sequence analysis.

BCVA by individual is shown in Figure 3. Six eyes retained normal vision 5 years after SRT. Eight of the 14 patients with 5 years of follow-up retained BCVA $\geq 6/12$. BCVA deteriorated to $\leq 20/200$ in four retained eyes.

Table 1 records the worst grade of any toxicity that was observed for each patient during follow-up. Retinopathy developed in 12 eyes. The CTCv4.0 grade (G) was G1 in six (three of which resolved), G2 in two, G3 in two, and G4 in two eyes. There was no optic neuropathy (ONp) without retinopathy. There was G1 ONp in one eye, while two cases of optic-disc oedema resolved. One case with G4 retinopathy secondary to central retinal vein occlusion also developed G4 ONp, glaucoma, and vitreous haemorrhage. Unexpectedly, some cases of grade 1 retinopathy and optic-disc oedema resolved. That is, at subsequent clinical examination, there was no longer any evidence of the previously observed toxicity, which was then scored as absent. Until now, one early cataract has been observed.

Discussion

SRT is an established radiation treatment modality for posterior CM [1]. Using dose fractionation regimens adopted from PBT, tumour control in the eye is reliably high ($\geq 90\%$), but the EQD is too high for normal tissues to tolerate. This leads to high rates of vision loss (90–100%), neovascular glaucoma (15–20%), and enucleation (10–20%) [6, 12]. Apart from cataract, vision loss after radiotherapy is multifactorial and may be due to any or all of the following: (i) radiation retinopathy and its complications, (ii) radiation ONp, (iii) tumour-related factors that are poorly described but probably include a prolonged local inflammatory response [13]. For treatment of tumours in close proximity to the ON and or macula, there is a balance between use of an adequate dose to achieve a high rate of tumour control and a dose that spares normal tissue sufficiently to allow for a chance of vision preservation.

This pilot study was designed to explore the hypothesis that the combination of a reduced EQD with standard

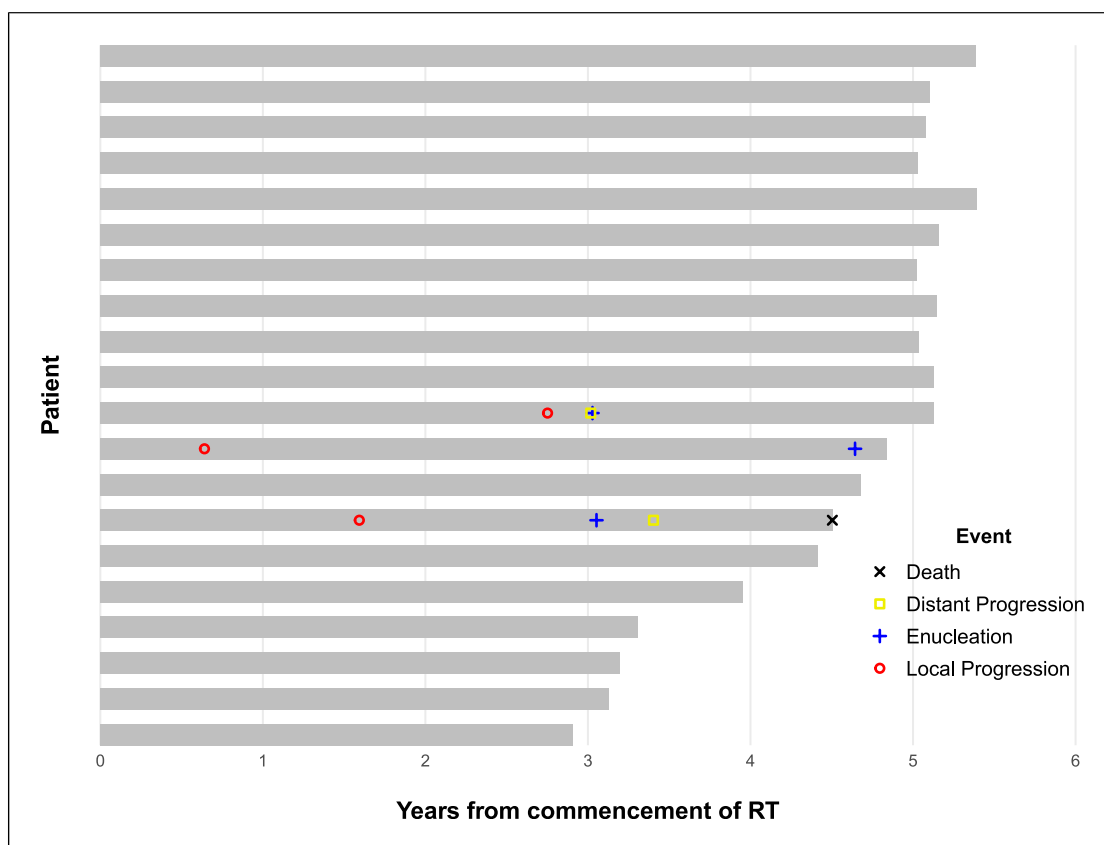


Fig. 2. Disease and enucleation events.

fractionation could preserve useful vision after treatment of peripapillary tumours, without compromising metastasis-free survival. With mature follow-up, 5-year MFS in this small cohort is 88% which is within expectation for T1-T2 CM [14]. FFLP in our cohort was acceptable at 85% at 5 years, although lower than >90%, which is expected after treatment with ablative dose fractionation regimens [1, 2]. The confidence intervals are wide in this small study, but a reduction in FFLP is not unexpected, given the EQD reduction. Until now, the FFLP achieved has not adversely affected MFS survival. A larger study is required to more accurately estimate FFLP and MFS after treatment with this fractionation which represents a significant dose reduction. Future questions include the relative importance of tumour volume versus tumour biology with regard to disease outcomes. The ultimate goal would be to gain a better understanding of case selection for safe use of a treatment which offers patients the possibility of vision preservation.

Thirteen patients retained a BCVA >6/60. More striking is that 5 years after SRT, 8 patients retained BCVA ≥6/12, five of whom retained normal vision. These results support the underlying hypothesis that preser-

vation of vision, at least to 5 years, is possible with standard fractionation combined with a reduced EQD.

The EQD is a complex entity which is affected by cellular repair and other parameters beyond the scope of this discussion [12, 15, 16]. It does not apply to plaque brachytherapy and cannot be used to directly compare between protons with photons. Nonetheless, it is a useful tool to compare dose fractionation regimens, and examples are shown in Table 2. The only phase III study of dose reduction for CM is the Boston study [5]. Patients were randomised between 50CGy and 70CGyE, both in five fractions. Five-year rates of tumour control and late toxicity were equivalent despite a >40% reduction in EQD. A randomised phase II study of protons for large CM also found no difference in FFLP or enucleation rate if 52 Gy was delivered in four or eight fractions [12]. The Vienna group systematically reduced the SRT dose from 70 Gy to 60 Gy and then 50 Gy in five fractions and found no difference in the rate of disease control in the eye, enucleation, or metastatic disease [2]. Most SRS centres deliver 30–50 Gy in 1 fraction. Shirmer et al. [17] reported a crude tumour control rate of 93% after SRS to 22.2 Gy.

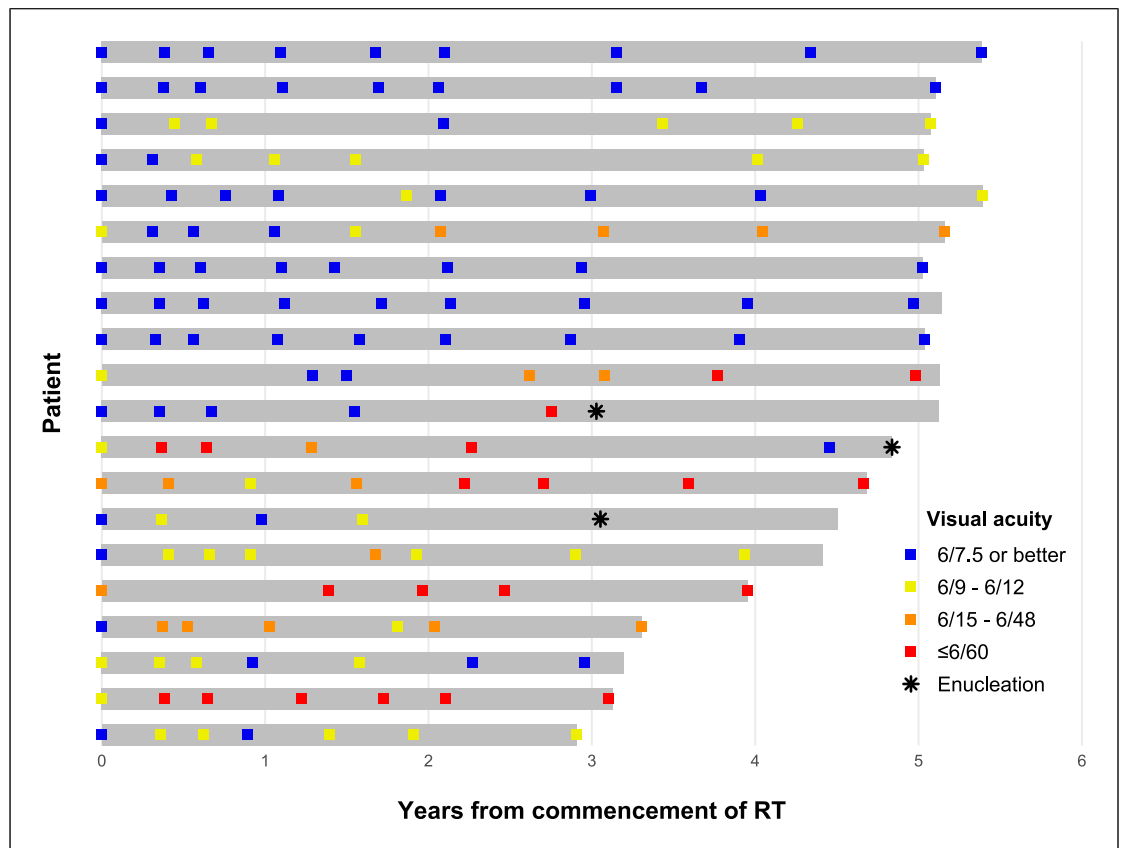


Fig. 3. BCVA.

Table 2. EQDs of common dose fractionation regimens

Prescription	Dose/fx	EQD, a/B = 2	EQD, a/B = 10	% change from 50 in 5	
				a/B = 2	a/B = 10
70 in 5	14 Gy	280	140	+87	+68
60 in 4	15 Gy	255	125	+70	+51
50 in 5	10 Gy	150	83	–	–
60 in 30	2 Gy	60	60	–60	–28

This is a calculated value that estimates the effect of different dose fractionation regimens on tumour and tissues. The a/B ratio is an estimate of cellular repair capacity. Classically, a/B for endothelium and nerves = 2 and for tumours = 10+, but slow growing tumours can have a low a/B [16]. The a/B of CM is unknown [17]. EQD, equieffective dose.

Several centres have explored brachytherapy dose reduction. Saconn et al. [18] reviewed the outcome of 62 patients treated with COMS plaque to deliver a mean apical dose of 62.5 Gy. They found similar disease control to the COMS study cohort and a suggestion of improved VA at 5 years. More recently, De Caluwé and colleagues assessed tumour control in the eye by minimum tumour dose after strontium-90 plaque. They found a minimum

dose of at least 65 Gy to be required for durable tumour control [19]. Altogether, this limited evidence of dose reduction for CM supports the idea that current regimens deliver a higher, more toxic dose than is needed for successful treatment of many CMs.

The relationship between FFLP and development of DM from CM is complex and not fully described [20–23]. There is the potential for DM to develop after treatment if

the primary tumour is not fully destroyed, while the propensity for primary tumour progression may be a marker of more aggressive disease that has already metastasised prior to local therapy. The Collaborative Ocular Melanoma Studies were conducted prior to tumour genotyping without stratification for this major determinant of risk for metastatic disease. The Ophthalmic Oncology Task Force pooled tumour, treatment, and survival data from 10 international ocular oncology centres. A total of 3,809 uveal melanoma entries were reviewed of which 3,217 iris and CM patients were included. In this large non-randomised population, local recurrence and extrascleral extension were associated with an increased risk of DM, whereas TNM stage was not [20]. More recently, it has been demonstrated that while larger tumours are more likely to harbour high metastatic-risk gene mutations, it is possible for low TNM stage tumours to be of high-risk genotype [24, 25]. It is also recognised that biologically good prognosis tumours may evolve over time to acquire a more aggressive phenotype [26]. A weakness of the current study is the absence of tumour genotyping before treatment as it is technically challenging to obtain tumour tissue from small, posterior tumours. There is growing evidence that genetic profile does not influence local control after ablative-dose radiotherapy [27]. It is not known if genetic profile affects outcome after the standard fraction reduced EQD radiotherapy. This will need to be addressed in the design of future studies of altered EQD for CM.

Other weaknesses of this study are its small size and the use of US rather than MRI for liver surveillance as a low-cost intervention. MRI liver is recognised as a more sensitive imaging modality for detection of metastases. Use of US rather than MRI may have affected the time to detection of liver metastases.

Conclusion

This pilot study is the first to report that it is possible to successfully treat posterior CM involving the ON, macula, or both with standard fraction SRT to 60 Gy and retain useful vision up to 5 years after treatment. This dose fractionation regimen warrants further investigation.

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Dr. William Campbell was not available to confirm co-authorship, but the corresponding author Dr. Claire E Phillips affirms that author Dr. William Campbell contributed to the paper, had the opportunity to review the final version to be published, and guarantees author Dr. William Campbell co-authorship status and the accuracy of the author contribution and conflict of interest statements.

Statement of Ethics

Institutional Ethics Committee approval was obtained at the Peter MacCallum Cancer Centre (Study 14/71), and the study was registered on the Australian New Zealand Clinical Trials Registry (ANZCTR) (12614000531617). All participants gave written informed consent before any study procedures.

Conflict of Interest Statement

No conflicting relationship exists for any author.

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Author Contributions

Claire Phillips contributed to protocol conception, writing the protocol, treatment of patients and follow-up, and manuscript writing. Arkan Youssef, junior doctor, assisted with data quality. Joseph Sia treated patients and assisted with trial follow-up and contributed to manuscript writing. Rod O'Day, Daniel McKay, and Fred Chen contributed to clinical management, follow-up, and manuscript writing. John D. McKenzie and William Campbell contributed to the original protocol design, clinical management, and follow-up. Mathis Bressel, statistician, contributed to design and analysis.

Data Availability Statement

All disease and VA data generated or analysed during this study are included in this article. A later manuscript will outline details of the radiotherapy dose-volume analysis (for a radiation oncology audience). Further enquiries can be directed to the corresponding author.

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